Data imputation and Animated visualization of Sound Pressure Level in Tallinn based on IoT Sensor Data

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- Figure 1 - Tallinn Smart City Sensors.

Introduction

Background

Urban Noise Challenges in Tallinn

Tallinn, as a rapidly developing European capital and a former European Green Capital, faces a complex acoustic environment. Urban noise is no longer viewed merely as an annoyance but as a critical environmental pollutant that impacts urban livability and real estate development. According to recent data from Statistics Estonia, approximately 40% of the city's population (roughly 160,000 residents) is currently exposed to traffic noise levels that exceed recommended safety thresholds (Tallinn City Government, 2024).

High Sound Pressure Levels (SPL), reaching peaks of 75 dB along major transit arteries such as Liivalaia and Pärnu mnt, significantly exceed the 55 dB night-time limit set for residential tranquility (ERR, 2025). This discrepancy has led to tangible consequences for city growth, including the denial of building permits in central districts where noise levels fail to meet legal safety standards (ERR, 2025).

The Evolution of Acoustic Monitoring and the IoT Shift

Historically, Tallinn relied on static noise mapping—mathematical simulations derived from infrequent manual measurements or outdated datasets (e.g., from 2019). However, the city is transitioning toward dynamic, real-time monitoring via IoT (Internet of Things) sensor networks. Supported by initiatives like "Test in Tallinn," this shift aims to provide a high-resolution, minutely understanding of how noise pulses through the city (Green Tallinn, 2024).

Figure 1. - Tallinn Smart City Sensors.

While IoT networks offer unprecedented granularity, they introduce significant technical vulnerabilities. Unlike high-grade industrial stations, low-cost IoT sensors are prone to periodic "missingness" caused by network packet loss (LoRaWAN/WiFi drops), power failures, or environmental interference (Smart Cities World, 2025). These gaps prevent the city from achieving a truly continuous and reliable acoustic profile.

The Critical Role of Data Imputation

Proper and correct analysis of Tallinn's noise levels is impossible if the underlying data is fragmented. If a sensor fails during a peak traffic hour, a simple average of the remaining data would mathematically underestimate the true noise pollution. To mitigate this, data imputation becomes a prerequisite for any meaningful urban analysis.

By implementing and optimizing a hybrid approach—combining K-Nearest Neighbors (KNN) for spatial gaps, Self-Imputation for historical patterns, and Google’s TimesFM for complex temporal sequences—this research aims to restore the integrity of the dataset. A "gap-free" dataset allows for more accurate Root Mean Square Error validation and ensures that noise-reduction policies are based on complete information rather than statistical artifacts.

Visual Analysis as a Diagnostic and Planning Tool

Beyond statistical recovery, the visual analysis of SPL data serves as the bridge between raw data and urban planning. Static tables of decibel values are difficult for stakeholders to interpret; however, animated visualization reveals the temporal flow of noise, making patterns like "noise waves" or "rush-hour pulses" visible and intuitive (Siigur, 2025).

Furthermore, the visualization tool serves a dual purpose as a diagnostic interface:

- Identification of Faulty Sensors: By comparing animated trends, analysts can visually spot sensors that are "stuck" or reporting impossible values.
- Mapping Uncertainty: By explicitly marking imputed data and uncertainty levels, the tool communicates the reliability of the analysis to the user.
- Strategic Expansion: Areas where the visualization shows high uncertainty or "blind spots" serve as direct recommendations for where more sensors should be placed to optimize the city's monitoring infrastructure.

Problem Statement

Unit of Study

The unit of study for this research is the **IoT-based acoustic monitoring network of Tallinn**, specifically focusing on the raw **Sound Pressure Level (SPL)** data streams generated by low-cost sensors deployed across major transit arteries. The study examines the data at a minutely resolution to capture the high-frequency temporal dynamics of the urban soundscape.

Motivation

The motivation for this research stems from the critical gap between Tallinn’s "Smart City" ambitions and the current unreliability of its environmental data.

- Policy Impact: Inaccurate noise mapping currently hinders urban development, leading to the denial of building permits based on outdated 2019 static simulations (ERR, 2025).
- Technical Necessity: Existing IoT networks suffer from periodic missingness due to hardware and connectivity failures. Without a specialized imputation framework, the resulting "dirty" data leads to mathematically skewed averages, making it impossible to accurately identify the city's most noise-polluted regions or verify the success of noise-reduction strategies.

Research Goal and Questions

Research Goal: The primary goal is to develop and validate a hybrid spatiotemporal imputation pipeline and an interactive visual analysis tool that transforms fragmented IoT data into a continuous, animated, and interpretable acoustic profile for Tallinn.

Research Questions:

- RQ1: To what extent does a hybrid approach (Self-Imputation, KNN, and TimesFM) reduce the Root Mean Square Error (RMSE) in reconstructed SPL datasets compared to traditional linear interpolation?
- RQ2: How can animated visual analysis be utilized to detect systemic failures, such as broken sensors or biased data, that are invisible in static reports?
- RQ3: In what ways does visualizing imputation uncertainty allow urban planners to identify geographical "blind spots" for future sensor deployment?

Relevant Concepts and Theory

This research is grounded in the following key concepts and existing theoretical frameworks:

- Sound Pressure Level (SPL): The logarithmic measure of the effective pressure of a sound relative to a reference value, measured in decibels (dB).
- Data Imputation (Missingness): The statistical process of replacing missing data with substituted values. This research builds on Spatiotemporal Correlation Theory, which posits that noise at a specific location is correlated both to its own historical patterns (Temporal) and to the readings of its geographic neighbors (Spatial).
- Foundation Models for Time-Series: The study utilizes Google TimesFM, a state-of-the-art decoder-only transformer model. Unlike traditional models, it leverages "Zero-Shot" learning to predict complex patterns in time-series data without requiring extensive local training.
- Visual Analytics: A field of data science that focuses on analytical reasoning supported by interactive visual interfaces. This research applies Temporal Heatmap Animation to translate raw dB values into intuitive, moving maps for non-technical stakeholders.

Research Methodology

Research Implementation and Methodology

To address the challenges of noise monitoring in Tallinn, this research implements a two-phase technical workflow:

1. Data Reconstruction Phase: A multi-model imputation pipeline is developed in Python to process raw IoT Sound Pressure Level (SPL) data. This involves the systematic identification of missing patterns and the application of an optimized mixture of models to reconstruct a high-fidelity dataset.
2. Analytical Visualization Phase: An interactive React-based dashboard is developed to consume the processed data. This system generates animated noise maps, enabling the

visual identification of temporal trends, the detection of faulty sensors, and the determination of areas where additional monitoring nodes are required.

Selection of Methods

The study utilizes a Hybrid Quantitative (Amina Ahmed, Lucas Pereira and Kimberly Jane, 2024) and Visual Analytic approach (G Kokk and S Jönsson, 2013). The technical framework consists of three distinct imputation layers:

- Self-Imputation (using historical): This method fills data gaps by analyzing an individual sensor's historical "seasonal" patterns, comparing specific timestamps (e.g., Tuesday at 08:00) against previous recorded periods to maintain temporal consistency.
- Weighted K-Nearest Neighbors (KNN): This technique leverages spatial correlation. By utilizing real-time readings from geographically adjacent sensors, weighted by the inverse of their distance (Weight = 1/Distance), the system estimates missing values based on the local acoustic environment.
- Google TimesFM (Foundation Model): A pre-trained Time Series Foundation Model handles long-duration gaps. This provides a learning capability that manages complex sequences where simple historical or spatial patterns may prove insufficient.

The efficency of these methods is validated through Synthetic Gap Analysis. By intentionally removing a subset of known data, the Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) are calculated to determine the optimal mixture of models required to approach the "ground truth."

Justification of the Proposed Framework

The proposed methodology is considered adequate for the Tallinn urban environment for the following reasons:

- Resilience to IoT Volatility: By integrating spatial (KNN), temporal (Self-Imputation), and deep-learning (TimesFM) methods, the framework ensures that the analysis remains robust even during significant network outages or system failures.
- Advanced Pattern Recognition: The inclusion of a Foundation Model is superior to traditional linear interpolation, as it accounts for the non-linear changes of urban noise caused by irregular events, such as roadworks or seasonal traffic shifts.
- Actionability through Visualization: While raw SPL data is dense and difficult to interpret, animated visual analysis provides a necessary layer of clarity for urban planning. It allows for the immediate recognition of changes and sensor drift, fulfilling the requirement for a proper and correct analysis to support the city's strategic noise reduction goals.

Validation of the Results

Validation Methodology: Synthetic Gap Testing

The primary method for validating the imputation algorithm is Synthetic Gap Testing. This involves taking a portion of the Tallinn IoT dataset where all values are known and intentionally removing a percentage of that data at random (e.g., 10% or 20%). By creating these "artificial gaps," a ground truth is established that the algorithm must attempt to reconstruct.

Defining the Baseline

To measure the effectiveness of the proposed hybrid model, a Baseline must be defined. The results of the hybrid algorithm (Self-Imputation + KNN + TimesFM) will be compared against:

- Linear Interpolation: The standard method of drawing a straight line between two points to fill a gap. If the hybrid algorithm produces values closer to the original removed data than the linear baseline does, the method is validated as superior.

Performance Metrics

The accuracy of the reconstruction will be measured using two standard mathematical formulas:

1. Mean Absolute Error (MAE): This measures the average distance between the imputed value and the real value in decibels (dB).
2. Root Mean Square Error (RMSE): This penalizes larger errors more heavily, which is critical for noise monitoring where a 5 dB mistake is much more significant than a 1 dB mistake.

Answering the Research Questions

This validation directly answers the research goals:

- Mathematical Proof: The RMSE and MAE scores prove if the data is accurate enough for city planning.
- Visual Proof: By using the React frontend, the reconstructed map can be visually compared to the original map. If the animated patterns (the noise pulses) look identical to the original, the visual analytic tool is considered reliable for identifying urban noise trends.

How to interpret the results:

- If the error is low: The algorithm is ready to be used on real unknown gaps in the Tallinn network.
- If the error is high in specific areas: It indicates where the city needs more sensors because the current neighbors (KNN) are too far away to provide a good estimate.

Summary of the Problem

Therefore, the central problem addressed by this thesis is the lack of a validated, hybrid imputation pipeline - utilizing KNN, Self-Imputation, and Google TimesFM - and an interactive visual analysis tool. Such a system is required to transform fragmented SPL IoT data into a continuous, animated, and interpretable acoustic profile for the city of Tallinn.

Literature Review

- **Acoustic Monitoring:** Current state of IoT sound sensors and Sound Pressure Level (SPL) standards.
- **Data Imputation Techniques:** Review of methods ranging from simple interpolation to Machine Learning (K-Nearest Neighbors, MICE, or RNNs).
- **Geospatial Visualization:** The role of Time-Series animation and Heatmaps in urban data science.
- **The Tallinn Context:** Existing smart city infrastructure in Estonia.

Methodology & Data Acquisition

- **Data Source:** Description of the Tallinn IoT sensor network (locations, hardware specifications, sampling frequency).
- **Data Pre-processing:** Cleaning raw decibel data, handling outliers, and synchronizing timestamps.
- **Imputation Framework:** * Defining the "Missingness" (Missing Completely at Random vs. Systematic gaps).
    - Selection of models for testing.
- **Visualization Stack:** Tools used (e.g., Python's Matplotlib/Plotly, Deck.gl, or QGIS Time Manager).

Implementation & Results

- **Imputation Performance:** Use metrics like Root Mean Square Error (RMSE) or Mean Absolute Error (MAE) to compare the accuracy of your filled-in data.
- **The Animation Pipeline:** How the temporal data was mapped to a visual coordinate system (x, y, t).
- **Case Studies:** Visualizing specific events (e.g., morning rush hour in Ülemiste, or a quiet night in Kalamaja).

Discussion

- **Analysis:** Did the imputation accurately capture the "rhythm" of the city?
- **Visual Impact:** How animation reveals patterns that static maps miss (e.g., the movement of noise "waves").
- **Limitations:** Connectivity issues, sensor calibration drift, or computational heavy-lifting.

Conclusion & Future Work

- **Summary:** Key findings regarding data reliability and visualization effectiveness.
- **Recommendations:** How the Tallinn Transport Department or City Planning could use this tool.
- **Future Work:** Integrating real-time traffic flow data or weather conditions.

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